```
SET SQL_SAFE_UPDATES = 0;



set @mean_warehousetohome=(select round(avg(warehousetohome))
from customer_churn where warehousetohome is not null);
update customer_churn
set warehousetohome=@mean_warehousetohome
where warehousetohome is null;
select * from customer_churn where warehousetohome is null;

select hourspendonapp,count(*) from customer_churn
where hourspendonapp is null group by hourspendonapp;
set @mean_hourspendonapp=(select round(avg(hourspendonapp))
from customer_churn where hourspendonapp is not null);
update customer_churn
set hourspendonapp= @mean_hourspendonapp
where hourspendonapp is null;
select * from customer_churn
where hourspendonapp is null;



set @mean_orderamounthikefromlastyear=(select round(avg(orderamounthikefromlastyear))
 from customer_churn
 where orderamounthikefromlastyear is not null);
update customer_churn
set orderamounthikefromlastyear=@mean_orderamounthikefromlastyear
```

```
where orderamounthikefromlastyear is null;

select * from customer_churn

where orderamounthikefromlastyear is null;


set @mean_daysincelastorder=(select round(avg(daysincelastorder)) from customer_churn

where daysincelastorder is not null);

update customer_churn

set daysincelastorder=@mean_daysincelastorder

where daysincelastorder is null;

select * from customer_churn where daysincelastorder is null;


set @mode_tenure = (select Tenure from customer_churn where Tenure is not null group by Tenure order by count(*) desc limit 1);

set @mode_coupon = (select CouponUsed from customer_churn where CouponUsed is not null group by CouponUsed order by count(*) desc limit 1);

set @mode_order  = (select OrderCount from customer_churn where OrderCount is not null group by OrderCount order by count(*) desc limit 1);


update customer_churn

set tenure=coalesce(tenure,@mode_tenure),

couponused =coalesce(couponused,@mode_coupon),

ordercount=coalesce(ordercount,@mode_ordercount)

where tenure or couponused or ordercount is null;

select * from customer_churn where tenure or ordercount or couponused is null;



delete from customer_churn where customerid>0 and warehousetohome>100;
```

```
update customer_churn
set preferredlogindevice="Mobile Phone"
where preferredlogindevice="Phone";

update customer_churn
set preferedordercat="Mobile Phone"
where preferedordercat="Mobile";

update customer_churn
set preferredpaymentmode="Cash on delivery"
where preferredpaymentmode="COD";

update customer_churn
set preferredpaymentmode="Credit Card"
where preferredpaymentmode="CC";


alter table customer_churn
rename column preferredordercat to PreferredOrderCat;

alter table customer_churn
rename column hoursspendonapp to HoursSpendOnApp;

alter table customer_churn
add column comlaintrecieved varchar(10);
```

```
update customer_churn
set comlaintrecieved=if(complain=1,"Yes","No");
select * from customer_churn;

alter table customer_churn
add column churnstatus varchar(10);
update customer_churn
set churnstatus=if(churn=1,"Churned","Active");
select * from customer_churn;
select distinct churnstatus from customer_churn;

alter table customer_churn
drop column churn,
drop column complain;

select * from customer_churn;

select churnstatus,count(*)
 from customer_churn
 group by churnstatus;
select round(avg(tenure)) as averagetenure,sum(cashbackamount) as totalcashbackamount
from customer_churn where churnstatus="Churned";
select round(avg(comlaintrecieved="Yes")*100,2) as percentage
from customer_churn where churnstatus="Churned";
select citytier, count(*) as churnedcustomers from customer_churn
where churnstatus="Churned" and preferredordercat="Laptop & Accessory"
```

```
group by citytier order by churnedcustomers desc limit 1;
select preferredpaymentmode,count(*) as activepaymentmode
from customer_churn where churnstatus="Active"
group by preferredpaymentmode order by count(*) desc limit 1;
select sum(orderamounthikefromlastyear) as singleandmobileorder
from customer_churn where maritalstatus="Single"
and preferredordercat="Mobile Phone";
select round(avg(NumberOfDeviceRegistered)) as upidevices
from customer_churn where PreferredPaymentMode ='UPI';
select citytier,count(*) as numberofcustomers
from customer_churn group by citytier
order by numberofcustomers desc limit 1;
select gender,sum(couponused) as totalcouponused from customer_churn
group by gender order by totalcouponused desc;
select PreferredOrderCat,COUNT(*)  as CustomerCount,
max(HoursSpendOnApp) as maximumhoursspent
from customer_churn group by PreferredOrderCat ;
select * from customer_churn;
select sum(ordercount) as totalorder
from customer_churn where preferredpaymentmode="Credit Card"
and satisfactionscore=(select max(satisfactionscore)
from customer_churn);
select round(avg(satisfactionscore))as complaintsatisficationscore
from customer_churn where comlaintrecieved="Yes";
select * from customer_churn;
select PreferredOrderCat , count(*) CustomerCount
```

```
from customer_churn where CouponUsed > 5 group by PreferredOrderCat
order by CustomerCount desc;
select preferredordercat,sum(cashbackamount) as cashbackamount
from customer_churn group by preferredordercat
order by cashbackamount desc limit 3;
select  PreferredPaymentMode,  count(*) CustomerCount  from customer_churn
where Tenure = 10 AND OrderCount > ((select sum(OrderCount)  from customer_churn) > 500)
group by PreferredPaymentMode order by CustomerCount desc ;
alter table customer_churn
add column DistanceCategory varchar(20);
update customer_churn
set DistanceCategory =
case
        when WarehouseToHome <= 5 then 'Very Close Distance'
        when WarehouseToHome <= 10 then 'Close Distance'
        when WarehouseToHome <= 15 then 'Moderate Distance'
else 'Far Distance'
end;
select DistanceCategory, ChurnStatus, count(*) as Count
from customer_churn
group by DistanceCategory, ChurnStatus;
select * from customer_churn
where maritalstatus="Married" and citytier=1 and
ordercount>(select avg(ordercount) from customer_churn);
create table customer_returns(
ReturnID int primary key,
```

```
CustomerID int ,
ReturnDate date,
RefundAmount int);
insert into customer_returns
values  (1001, 50022, '2023-01-01', 2130),
                (1002, 50316, '2023-01-23', 2000),
                (1003, 51099, '2023-02-14', 2290),
                (1004, 52321, '2023-03-08', 2510),
                (1005, 52928, '2023-03-20', 3000),
                (1006, 53749, '2023-04-17', 1740),
                (1007, 54206, '2023-04-21', 3250),
                (1008, 54838, '2023-04-30', 1990);
select * from customer_returns;
select cr.*,cc.* from customer_returns as cr
join customer_churn as cc on cr.CustomerID = cc.CustomerID
where  cc.ChurnStatus = 'Churned' and cc.ComlaintRecieved = 'Yes';
```